Charged particles

$$\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c} \ddot{\vec{d}}^{\,2} \sin^2\theta$$

radiated power

$$P = \frac{d\mathcal{E}_{rad}}{dt} = \int \frac{dP}{d\Omega} d\Omega = \frac{\mu_0 \ddot{\vec{d}}^{\,2}}{16\pi^2 c} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \sin^2\theta$$

$$\int_0^{\pi} d\theta \sin^3\theta = -\int_1^{-1} d(\cos\theta)\,(1-\cos^2\theta) = 2\int_0^1 dx\,(1-x^2) =$$

$$= 2\left(1-\frac{1}{3}\right) = \frac{4}{3}$$

$$\Rightarrow \quad P = \frac{\mu_0 \ddot{\vec{d}}^{\,2}}{16\pi^2 c} \frac{8\pi}{3} = \frac{\mu_0 \ddot{\vec{d}}^{\,2}}{6\pi c}$$

This applies to motion of charged particle as well — it radiates energy as long as $\ddot{\vec{d}} \neq 0$.

Consider a problem of particle, with charge q, colliding head on with static charge Q. The motion one-dimensional.

At spatial infinity $\vec{v} = v_0 \hat{r}$

$$\vec{d} = q\vec{r} \Rightarrow \ddot{\vec{d}} = q\vec{a}$$

Newton's 2nd law: $m\vec{a} = -\vec{\nabla} U$

$$ma = -\frac{dU}{dr}$$

$$\frac{d\mathcal{E}_{rad}}{dt} = \frac{d\mathcal{E}_{rad}}{dr}\frac{dr}{dt}$$

$$\mathcal{E}_{rad} = \int \frac{d\mathcal{E}_{rad}}{dt} dt = \int_{\infty}^{r_{min}} \frac{d\mathcal{E}_{rad}}{dr} dr$$

$$a = -\frac{1}{m}\frac{dU}{dr} \Rightarrow \vec{a}^{\infty} = -\frac{q}{m}\frac{dU}{dr}\,\hat{r}$$

$$\mathcal{E}_{rad} = \frac{\mu_0}{6\pi c}\int_{\infty}^{r_{min}} dr\,\frac{\left(\frac{q}{m}\frac{dU}{dr}\right)^2}{\frac{dr}{dt}}$$

Conservation of energy: $\frac{1}{2}m\left(\frac{dr}{dt}\right)^2 + U(r) = \frac{1}{2}mv_0^2$

$$\Rightarrow \frac{dr}{dt} = -\sqrt{v_0^2 - \frac{2U(r)}{m}}$$

here we assume that radiated energy is much smaller than $\frac{1}{2}mv_0^2$.

$$\Rightarrow \mathcal{E}_{rad} = \frac{\mu_0 q^2}{6\pi c m^2}\int_{r_{min}}^{\infty} dr\,\frac{\left(\frac{dU}{dr}\right)^2}{\sqrt{v_0^2 - \frac{2U(r)}{m}}}$$

And ~~the~~ r_{min} is determined from $v_0^2 - \frac{2}{m}U(r_{min}) = 0$.

Radiation Energy lost during such collision is $2\mathcal{E}_{rad}$:

$\mathcal{E}_{rad}$ on motion from $r=\infty$ to r_{min},

and another $\mathcal{E}_{rad}$ ~~from~~ $r_{min} \rightarrow r=\infty$.

Coulomb problem: $U(r) = \frac{qQ}{4\pi\varepsilon_0 r}$